Quadratic Forms over the Full Nimber Field

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Abstract

Let K be a perfect field of characteristic 2 for which the Artin–Schreier map $x \mapsto x^2 + x$ is surjective. Every nonsingular finite-dimensional quadratic form over K is hyperbolic, and therefore $W_q(K) = 0$. More generally, every quadratic form is classified by

$$\big(\dim V, \dim \operatorname{rad} B, \dim \operatorname{rad} Q\big),$$

where B is the polar form and $\operatorname{rad} Q = \{x \in \operatorname{rad} B : Q(x) = 0\}$. Conway's full field of nimbers On_2 is algebraically closed of characteristic 2, so the theorem applies to On_2: there is no transfinite Arf coordinate, and the Clifford class of every nonsingular form is split. Finite-field Arf invariants remain valid relative invariants over a declared finite nim-subfield, but scalar extension from $\mathbb{F}_{2^d}$ to $\mathbb{F}_{2^{2d}}$ kills them. We conclude by separating this full-field theorem from computations in a partial ordinal-nimber backend, which neither represents all of On_2 nor constructs every required root.

1 Introduction

Nim addition and nim multiplication make the class of ordinals into Conway's algebraically closed field On_2 of characteristic 2 [4, 8]. Quadratic-form invariants over finite nimber fields therefore raise a natural base-change question: what remains of the Arf invariant after extension to the full nimber field? The answer is that the nonsingular invariant vanishes. The singular theory retains only polar rank and one radical flag.

This conclusion is an application of standard characteristic-2 quadratic form theory [2, 3, 6]. Its role here is to identify the correct ground field and to separate two assertions that should not be conflated:

1. over full On_2, every required square root and Artin–Schreier root exists, so the classification is complete;
2. a finite implementation of ordinal nim arithmetic may classify a form over a certified finite subfield without being able to construct the isometry over On_2.

Set-sized quadratically closed initial segments of the nimbers have also been studied in [7, 9]; the theorem below requires only the field hypotheses stated explicitly and then uses algebraic closure for the full-On_2 corollary.

2 Quadratic forms in characteristic two

Let K be a field of characteristic 2 and V a finite-dimensional K-vector space. A quadratic form is a map $Q : V \to K$ satisfying

$$Q(av) = a^2 Q(v), \qquad B(u, v) = Q(u + v) + Q(u) + Q(v),$$

where the polar form B is bilinear. The form B is alternating. We put

$$R = \operatorname{rad} B = \{v : B(v,w) = 0 \text{ for all } w \in V\}, \qquad \operatorname{rad} Q = \{r \in R : Q(r) = 0\}.$$

The quadratic form is *nonsingular* when $R = 0$. This is the standard characteristic-2 convention [6, Chapter II, Section 7].

Write

$$\wp : K \longrightarrow K, \qquad \wp(t) = t^2 + t.$$

For a nonsingular form and a symplectic basis $(e_i, f_i)_{i=1}^r$, its Arf class is represented by

$$\sum_{i=1}^{r} Q(e_i)Q(f_i) \pmod{\wp(K)}.$$

Over a finite field, absolute trace identifies $K/\wp(K)$ with $\mathbb{F}_2$. Over a field on which $\wp$ is surjective, the quotient is zero.

Let H denote the hyperbolic plane, with basis e, f satisfying

$$Q(e) = Q(f) = 0, \qquad B(e,f) = 1.$$

Lemma 2.1 (Splitting a symplectic plane)**.** *If $\wp$ is surjective, every nonsingular quadratic plane over K is hyperbolic.*

Proof. Choose e, f with $B(e,f) = 1$ and write $a = Q(e)$ and $b = Q(f)$. If $a = 0$, then e is isotropic. If $a \neq 0$, choose t with $t^2 + t = ab$ and put $v = (t/a)e + f$. Then

$$Q(v) = \frac{t^2 + t + ab}{a} = 0.$$

Thus in either case there is a nonzero isotropic vector v. Choose w_0 with $B(v, w_0) = 1$ and set $w = w_0 + Q(w_0)v$. Since $Q(v) = 0$,

$$Q(w) = Q(w_0) + Q(w_0)B(w_0, v) = 0, \qquad B(v,w) = 1.$$

Hence (v, w) is a hyperbolic basis. □

Proposition 2.2. *If $\wp$ is surjective, every nonsingular quadratic space of dimension $2r$ is isometric to $H^{\perp r}$. Consequently $W_q(K) = 0$.*

Proof. A nondegenerate alternating form has a symplectic plane. By Lemma 2.1, the restriction of Q to that plane is hyperbolic. Its polar-orthogonal complement is nonsingular, so induction splits off all r planes. Every nonsingular form is therefore Witt-trivial. □

The remaining information lies on the polar radical. Perfection is used precisely here.

Lemma 2.3 (Zero-polar forms)**.** *Suppose K is perfect and R is an s-dimensional space on which the polar form of Q vanishes. There is a unique linear functional $\ell : R \to K$ such that $Q(r) = \ell(r)^2$. Up to isometry,*

$$Q|_R \cong \begin{cases} 0^{\perp s}, & \ell = 0, \\ \langle x^2 \rangle \perp 0^{\perp (s-1)}, & \ell \neq 0. \end{cases}$$

Proof. For a basis $r_1, \ldots, r_s$, perfection gives unique c_i with $c_i^2 = Q(r_i)$. Define $\ell(\sum_i x_i r_i) = \sum_i c_i x_i$. Since the polar form vanishes,

$$Q\left(\sum_i x_i r_i\right) = \sum_i x_i^2 Q(r_i) = \ell\left(\sum_i x_i r_i\right)^2.$$

Injectivity of Frobenius gives uniqueness. If $\ell \neq 0$, choose a basis in which ℓ is the first coordinate. □

3 Classification over an Artin–Schreier-surjective field

Theorem 3.1. *Let K be a perfect field of characteristic 2 with surjective Artin–Schreier map. Let Q be a quadratic form on V, let B be its polar form, and write*

$$\operatorname{rank} B = 2r, \qquad \dim \operatorname{rad} B = s.$$

Then exactly one of the following normal forms holds:

$$(V,Q) \cong \begin{cases} H^{\perp r} \perp 0^{\perp s}, & Q|_{\operatorname{rad} B} = 0, \\ H^{\perp r} \perp \langle x^2 \rangle \perp 0^{\perp (s-1)}, & Q|_{\operatorname{rad} B} \neq 0. \end{cases}$$

In the second case $s \geq 1$. Two such forms are isometric if and only if their triples

$$\big(\dim V, \dim \operatorname{rad} B, \dim \operatorname{rad} Q\big)$$

agree.

Proof. Choose a vector-space complement W to $R = \operatorname{rad} B$. The restriction $B|_W$ is nondegenerate: a vector in its radical is orthogonal to both W and R, hence lies in $W \cap R = 0$. Therefore

$$(V,Q) = (W, Q|_W) \perp (R, Q|_R).$$

Proposition 2.2 identifies the first summand with $H^{\perp r}$, and Lemma 2.3 gives the two radical summands. In the notation of that lemma, $\operatorname{rad} Q = \ker \ell$, so its dimension is s in the first case and $s-1$ in the second. These dimensions recover the normal form and are invariant under isometry. □

Corollary 3.2 (The full nimber field)**.** *Every nonsingular finite-dimensional quadratic form over* On_2 *is hyperbolic, and*

$$W_q(\mathrm{On}_2) = 0.$$

Every singular form over On_2 *has exactly one of the two normal forms in Theorem 3.1. In particular, no additional transfinite Arf invariant occurs.*

Proof. The field On_2 is algebraically closed of characteristic 2 [4, 8]. Algebraic closure makes both Frobenius and $x \mapsto x^2 + x$ surjective, so Theorem 3.1 applies. □

Remark 3.3 (Set-sized reduction)*.* Only finitely many coefficients and finitely many roots occur in the proof of a finite-dimensional instance. They lie in a set-sized algebraically closed subfield of On_2. Thus the corollary does not require performing linear algebra over a proper-class carrier.

4 Finite nim-subfields and the Arf invariant

Let $K_d = \mathbb{F}_{2^d} \subset \mathrm{On}_2$. For a nonsingular quadratic form over K_d, the Arf class lies in

$$K_d/\wp(K_d) \cong \mathbb{F}_2,$$

with the isomorphism given by absolute trace. This is a relative invariant: it depends on the ground field over which changes of basis are allowed.

Proposition 4.1. *Scalar extension from K_d to K_{2d} induces the zero map*

$$W_q(K_d) \longrightarrow W_q(K_{2d}).$$

Consequently the directed colimit of the finite-field quadratic Witt groups inside On_2 *is zero.*

Proof. For $c \in K_d$, the polynomial $X^2 + X + c$ either splits over K_d or is an irreducible quadratic, in which case it splits over K_{2d}. Thus every class in $K_d/\wp(K_d)$ becomes zero in $K_{2d}/\wp(K_{2d})$. The finite-field classification by the Arf class [2, 5] gives the Witt-group statement. Every element at every finite stage therefore dies at a later stage, which is exactly the vanishing of the colimit. □

For example, the plane with $Q(e) = Q(f) = B(e, f) = 1$ is anisotropic over $\mathbb{F}_2$. If $t^2 + t + 1 = 0$ in $\mathbb{F}_4$, then $te+f$ is isotropic, so the same plane is hyperbolic over $\mathbb{F}_4$. The original Arf bit correctly records failure to split over $\mathbb{F}_2$; it is not invariant under unrestricted scalar extension.

5 Clifford consequence

Corollary 5.1. *For every nonsingular finite-dimensional quadratic form Q over On_2, the graded Morita class of $\mathrm{Cl}(Q)$ is split on the characteristic-2 Clifford–Brauer–Wall surface.*

Proof. By Corollary 3.2, $Q \cong H^{\perp r}$. Orthogonal sum corresponds to graded tensor product of Clifford algebras [6, Chapter IX]. For a hyperbolic basis e, f,

$$e^2 = f^2 = 0, \qquad ef + fe = 1,$$

so $\mathrm{Cl}(H)$ is a graded matrix algebra and is graded Morita-trivial. Hence $[\mathrm{Cl}(Q)] = 0$. □

This statement concerns Clifford algebras of nonsingular quadratic forms. For singular Q, the full Clifford algebra need not be graded central simple, so the singular radical flag in Theorem 3.1 is quadratic-form data rather than a Brauer–Wall coordinate.

6 Executable and formal boundary

The mathematical theorem concerns full On_2. Ogdoad's executable `Ordinal` scalar is instead a partial, checked representation of an initial Kummer region. Its multiplication rejects computations that leave the verified window. For a finite metric, the routine `ordinal_common_finite_subfield_degree` returns a common finite degree only when the necessary excess data and checked degree arithmetic are available. The finite-field Arf and Brauer–Wall reports are then relative to that recorded degree, exactly as Proposition 4.1 requires.

None of these routines asserts that the represented scalar is algebraically closed. Nor do they promise to construct the square roots and Artin–Schreier roots used by Theorem 3.1. A failure to produce an in-window witness is therefore an implementation boundary, not a third mathematical isometry class.

The companion Lean development, distributed with Ogdoad [1], checks the full finite-dimensional theorem over a set-sized field. `Off.lean` proves the exact plane and radical lemmas. `SymplecticBasis.lean` constructs the recursive orthogonal basis. `CharTwoClassification.lean` assembles an explicit `QuadraticMap.IsometryEquiv` from the original form to the coordinate normal form. The formal hypotheses are exactly perfection and Artin–Schreier surjectivity. Every nondegenerate plane is hyperbolized within the same span, and the terminal polar radical is proved to be either $0^{\perp s}$ or $\langle x^2 \rangle \perp 0^{\perp(s-1)}$. The returned object contains every direct-sum witness and every basis change. It computes the dimensions of the polar and quadratic radicals of the coordinate form, proves those dimensions invariant under isometry, and kernel-checks the converse in `equivalent_iff_dimension_radicals_eq`; hence the final iff clause of Theorem 3.1 is formalized as stated. The algebraically closed theorem is an immediate specialization. The development does not encode Conway's proper-class field or identify Mathlib field operations with the partial Rust arithmetic.

7 Conclusion

Perfectness and Artin–Schreier surjectivity remove the nonsingular characteristic-2 obstruction completely. Over full On_2, every nonsingular quadratic form is hyperbolic, $W_q(\mathrm{On}_2) = 0$, and its associated Clifford class is split. A singular form retains only the dimensions of its polar and quadratic radicals. Finite-field Arf bits remain meaningful when their finite ground field is recorded, but they vanish after a quadratic field extension and hence in the directed limit. This field-theoretic classification is independent of whether a partial ordinal backend can construct the corresponding change of basis.

References

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